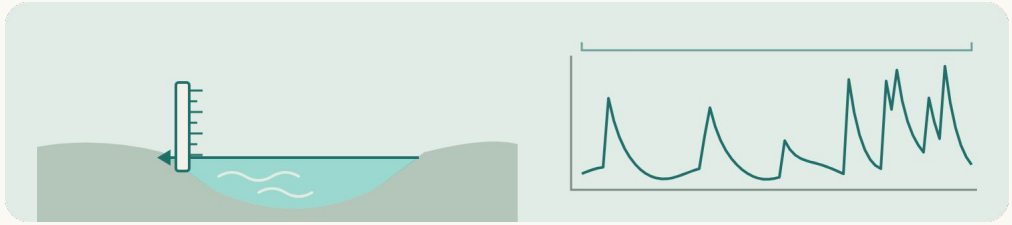
COM7022 · MACHINE LEARNING · REAL-WORLD CASE STUDY · September 2026

Predicting Floods with Machine Learning

Google Flood Hub forecasts river floods up to 7 days ahead.

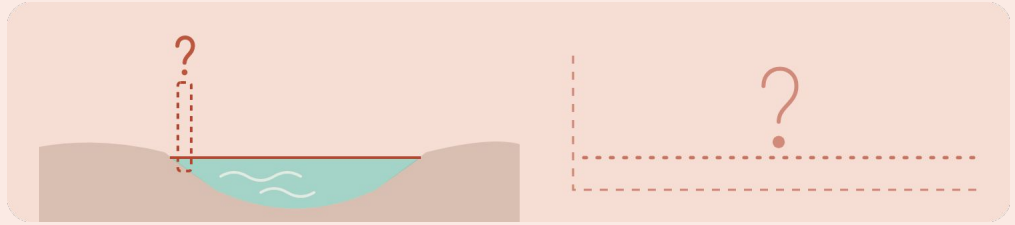
TEAM #2 Margaret Chebiwot Raymond Okyere Kaveh Momeni

The flood forecasting data gap



Gauged rivers

1. Stream gauge records river level
2. Long local history, often 30+ years
3. River-specific forecasting model
4. **Reliable local warning**



Ungauged rivers

1. No measuring station
2. No local water-level history
3. Traditional model lacks local evidence
4. **Warning gap**

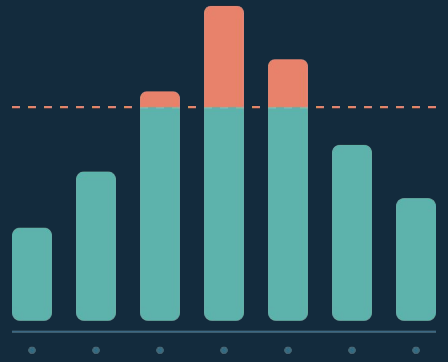
KEY MESSAGE AI addresses the gap by learning transferable patterns from many gauged watersheds.

Sources [1–3]

OPERATIONAL REACH

Flood Hub's operational reach

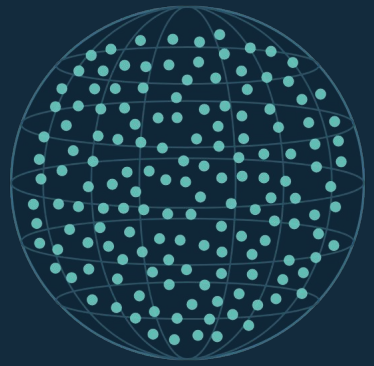
7



days ahead

Maximum forecast lead time

150+



countries

Geographic coverage

2B



people, approx.

Population in covered areas

Free



public access

Flood Hub is free to use

Sources [1, 2, 4, 5]

FORECAST PROCESS

How Flood Hub produces a forecast

Two models turn weather forecasts into river-flow estimates and flood maps.



INPUTS

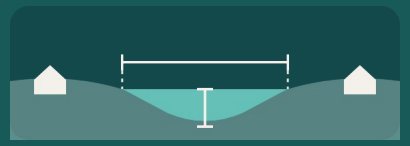
- Weather forecasts
- Rainfall & temperature
- Basin & soil attributes



STEP 1

Hydrological model

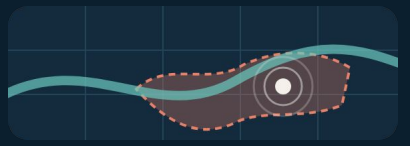
Estimates river flow and water level for each forecast horizon.



STEP 2

Inundation model

Combines predicted flow with terrain to estimate flood extent and depth.



OUTPUT

Flood map and risk information

For communities and authorities.

KEY MESSAGE

Training across gauged basins allows the model to transfer hydrological patterns to rivers with little or no local measurement data.

Sources [1, 2, 4, 5]

TRAINING SEQUENCE

Why this is supervised learning

Training pairs predictor variables with measured river flow, which provides a known target.

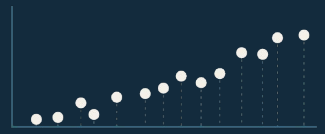
1



Predictors

Rainfall,
soil moisture &
basin properties

2



Labeled examples

Predictors paired
with observed
river flow

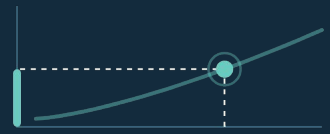
3



Model

Learns a mapping,
from conditions
to flow

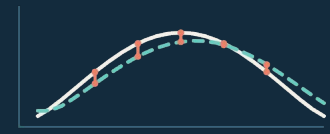
4



Prediction

Produces a
numeric river-flow
estimate

5



Error and update

Compares
prediction,
with observation

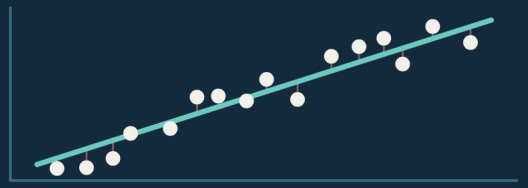
Sources [1, 2]

TAXONOMY

The machine-learning taxonomy

Three approaches; only one matches the primary forecasting objective.

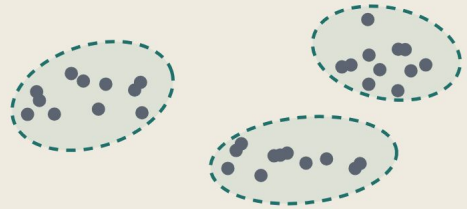
SELECTED APPROACH



Supervised learning

Known labels guide a numeric regression task. Each observed flow value provides a correction signal.

NOT SELECTED



Unsupervised learning

Finds structure without target labels. That does not match the primary forecasting objective.

NOT SELECTED



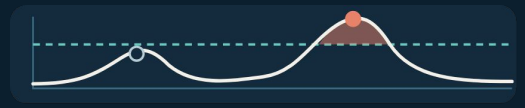
Reinforcement learning

Learns from actions and rewards. Flood Hub predicts hazards rather than controlling physical infrastructure.

Sources [1, 2]

ETHICS AND GOVERNANCE

Three decisions that need human ownership



1 Alert thresholds



2 Data inequality



3 Accountability

	Alert thresholds	Data inequality	Accountability
DECISION	When should a public warning trigger?	How should uncertainty be communicated?	Who carries responsibility when a forecast fails?
RISK	A low threshold creates false alarms. A high threshold can miss harmful events.	Sparse observations can reduce forecast confidence in vulnerable regions.	A private platform supports decisions with public-safety consequences.
SAFEGUARD	Local agencies set thresholds and monitor real-world performance.	Show uncertainty, validate locally and invest in regional observations.	Define roles, audit failures and maintain fallback warning channels.

Sources [1, 5, 6]

KEY MESSAGE

Prediction supports decisions.

Public authorities remain accountable for warnings and emergency response.



Sources [1, 5, 6]

SOURCES

References

1 Nearing, G. et al. (2024). *Global prediction of extreme floods in ungauged watersheds*. Nature, 627, 559–563. https://www.nature.com/articles/s41586-024-07145-1

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4 Google Blog (2024). *Expanding Flood Forecasting Coverage*. https://blog.google/innovation-and-ai/products/expanding-flood-forecasting-coverage-helping-partners/

5 Google Flood Hub Help. *How Does the Inundation Model Work?* https://support.google.com/flood-hub/answer/15636595?hl=en

6 Google Research. *Flood-Forecasting Code Repository*. https://github.com/google-research/flood-forecasting